Paper review · Empiricist

View, scene and instruction in π0.5-DROID

DROID, Khazatsky et al., RSS 2024 (arXiv:2403.12945).

Pavel Bushuyeu · November 2026

Input sensitivity of π0.5-DROID

The question

DROID argues that diverse scenes, tasks and camera viewpoints make policies generalize.

π0.5-DROID is Physical Intelligence's open π0.5, fine-tuned on DROID.

Without a robot: how much do its actions depend on the camera view, the scene and the instruction?

Hypotheses (set before the run)

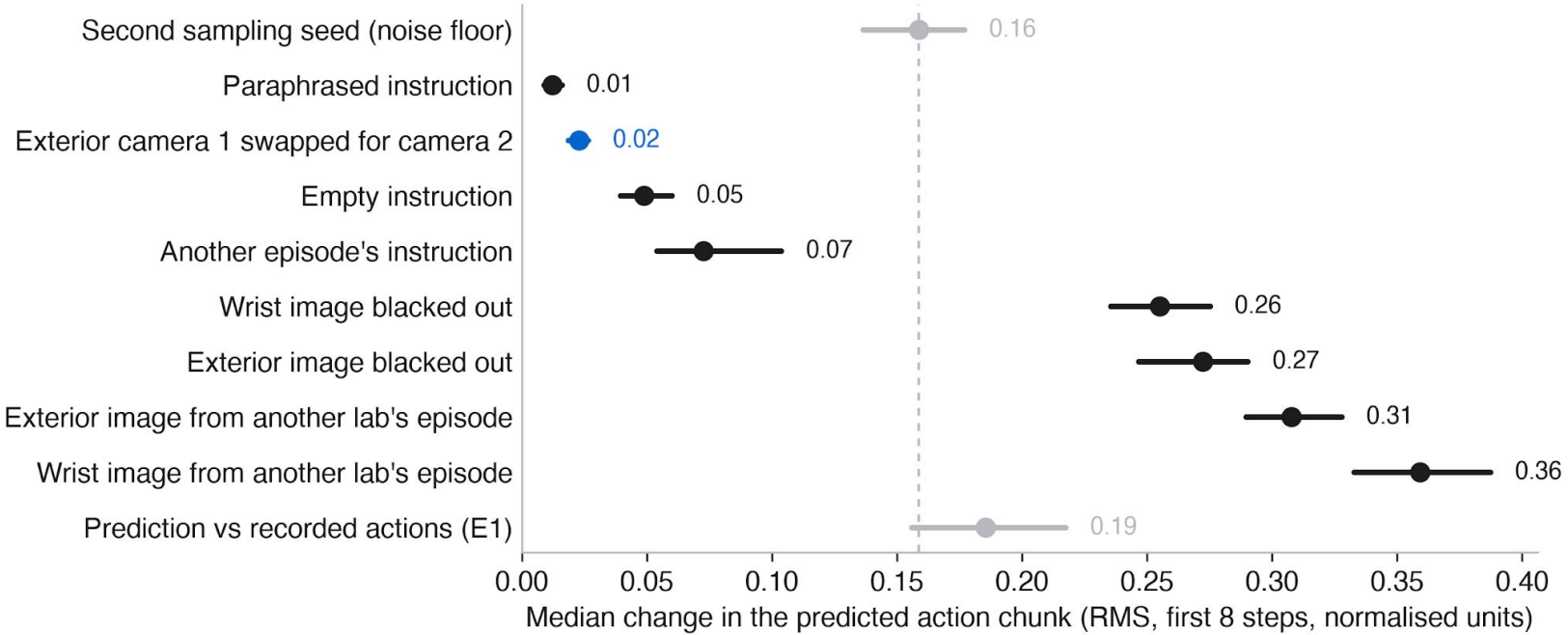
H1: camera diversity buys view invariance. Exterior camera 2 instead of camera 1 changes the actions about as much as a new sampling seed.

H2: language matters early in an episode. A wrong instruction moves the actions well above the noise early, and less late.

Offline probes on 500 DROID states

Model	pi05_droid (openpi): exterior + wrist image, state, instruction → 15 actions.
Data	droid_100: the first 100 DROID episodes, 13 labs; 5 states per episode.
Probes	Other camera, other or empty instruction, paraphrase, black or other-lab images.
Yardstick	Noise floor: the change from a second sampling seed. Probes share the seed.
Measure	RMS change over the first 8 actions, normalised units; medians, episode bootstrap.
Compute	RTX 3090 (iolani-5): 5,140 calls at 177 ms, about 30 min.

Action change by probe



View invariance with scene dependence

Camera 1 → camera 2

Same seed: change 0.023, 0.15× the noise floor [0.12, 0.16].

Above the floor in 1.4% of states; the same in every phase and in all 13 labs.

Accuracy against the recorded actions: no change.

Control: another lab's scene

Change 0.31: 1.9× the floor, 13× the swap.

Error against the recorded actions rises by 0.14.

H1 holds, more strongly than stated: the model reads the scene, not the viewpoint.

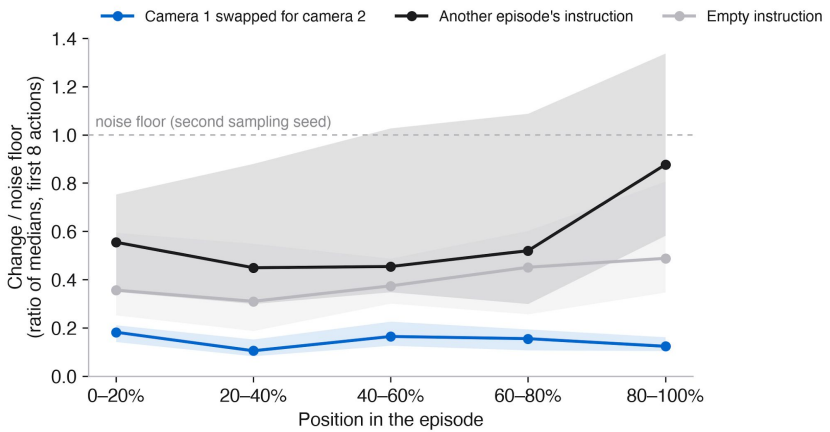
Language effect below sampling noise

Instruction	× noise floor
Another episode's	0.58 [0.43, 0.85]
Empty	0.39 [0.32, 0.48]
Paraphrase	0.09 [0.07, 0.12]

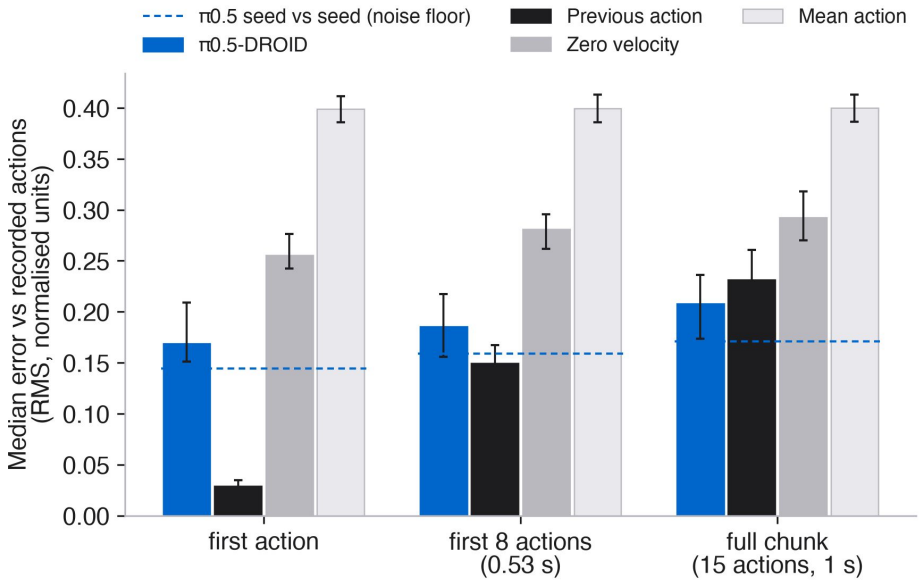
H2 fails: early 0.51, late 0.82 (difference not significant).

19% of states change by more than 2× the floor.

The true instruction still lowers the error.



Agreement with the recorded actions



Error 0.185 = 1.17× the noise floor: most of the open-loop error is the model's own sampling spread.

Repeating the previous action wins the first 8 steps; π0.5 wins over the full 1 s chunk.

Failed episodes: error 0.37; successful: 0.15.

Surprises in droid_100

What the sample holds

54 of 100 episodes have no instruction, including all 19 failures.

droid_100 is RLDS v1.0.0: instructions for about 30k episodes; v1.0.1 has about 75k (openpi).

One lab (TRI) has 29 of the 100 episodes.

What it means

Language probes use the 46 annotated episodes.

Use v1.0.1 for language work.

A 100-episode slice does not show the diversity of the full dataset.

What matches and what does not

Claim	DROID paper	This probe
Viewpoints	Viewpoint diversity helps with new camera angles; policies use both exterior cameras.	Matches, in-distribution: one view or the other gives the same actions.
Scenes	Scene diversity helps with new scenes; closed-loop tests in new scenes.	Consistent: actions depend on the scene (1.9–2.3×). Transfer not tested.
Language	86 tasks with instructions; openpi: "good language-following".	A wrong instruction moves actions less than sampling noise in DROID's scenes.

Scene and state set the action

Challenges

A shared GPU: the run waited 69 min for two TTS jobs.

Disk: 24 GiB at peak on a 95% full disk; all deleted after.

The checkpoint's training recipe is inferred from openpi's open loader.

Insights

Compare with the policy's own sampling noise: an error of 0.185 alone looks large, but it is 1.17× the noise.

In DROID's scenes, a wrong instruction rarely changes what the robot does next.

Held-out views, same-scene tasks

Hypothesis: the view invariance holds for camera poses outside DROID, and the language effect rises when two tasks are possible in one scene.

Views: new exterior camera poses on a DROID setup, then the same swap probe.

Language: pairs of feasible instructions in one scene (the cup into the bowl, or out of it).

Measure: the same ratios to the noise floor, then success in closed loop.

Sources

1. Khazatsky et al. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset, RSS 2024.
2. Physical Intelligence. π0.5: a Vision-Language-Action Model with Open-World Generalization, 2025.
3. Driess et al. Knowledge Insulating Vision-Language-Action Models, 2025.
4. Physical Intelligence. openpi: pi05_droid checkpoint and DROID training pipeline (commit 215abfb).
5. DROID team. droid_100 sample, RLDS v1.0.0, gs://gresearch/robotics/droid_100.